Available online at www.sciencedirect.com
ScienceDirect
Reliability Engineering and System Safety 94 (2009) 162–179
RELIABILITY ENGINEERING & SYSTEM SAFETY
www.elsevier.com/locate/ress

Design optimization of a safety-instrumented system based on RAMS + C addressing IEC 61508 requirements and diverse redundancy

A.C. Torres-Echeverría[a,*], S. Martorell[b], H.A. Thompson[a]

[a]Department of Automatic Control and System Engineering, The University of Sheffield, Mappin Street, Sheffield S1 3JD, UK
[b]Department of Chemical and Nuclear Engineering, Polytechnic University of Valencia, Camí de Vera sn, 46022 Valencia, Spain

Received 16 May 2007; received in revised form 7 February 2008; accepted 13 February 2008
Available online 10 March 2008

Abstract

This paper presents the design optimization by a multi-objective genetic algorithm of a safety-instrumented system based on RAMS + C measures. This includes optimization of safety and reliability measures plus lifecycle cost. Diverse redundancy is implemented as an option for redundancy allocation, and special attention is paid to its effect on common cause failure and the overall system objectives. The requirements for safety integrity established by the standard IEC 61508 are addressed, as well as the modelling detail required for this purpose. The problem is about reliability and redundancy allocation with diversity for a series–parallel system. The objectives to optimize are the average probability of failure on demand, which represents the system safety integrity, Spurious Trip Rate and Lifecycle Cost. The overall method is illustrated with a practical example from the chemical industry: a safety function against high pressure and temperature for a chemical reactor. In order to implement diversity, each subsystem is given the option of three different technologies, each technology with different reliability and diagnostic coverage characteristics. Finally, the optimization with diversity is compared against optimization without diversity.
© 2008 Elsevier Ltd. All rights reserved.

Keywords: Safety system; IEC 61508; Common cause failure; Diversity; Safety integrity level; Optimization; Genetic algorithm; Safety; Reliability

1. Introduction

Safety-instrumented systems (SIS) pertain to the generic category established by the international standard IEC 61508 [1] as safety-related systems. "Safety-instrumented system" is a term more currently used for process industry. It is defined by IEC 61511 [2], a standard derived from IEC 61508 specific for this sector, as an "*instrumented system used to implement one or more safety instrumented* functions. *A SIS is composed of any combination of sensors, logic solver and final elements*". The design of SIS entails achievement of some minimum levels of safety integrity, as required by IEC 61508. The safety integrity requirements include the restriction of the system average probability of failure on demand (PFD_{avg}) to a maximum target limit and the compliance with some minimum levels of fault tolerance. Fault tolerance, the capacity of a system to prevent single failures escalating into system failures, is usually achieved by some form of redundancy. The most basic forms of redundancy are hardware and software redundancy. The implementation of hardware redundancy implies the use of extra equipment or parts that the system would not normally need to perform its function, but are used to tolerate potential faults.

Plain identical hardware redundancy may, however, introduce some collateral issues that must not be overlooked, since that may lead to overoptimistic designs. Several identical redundant components are sensitive to stress factors that may lead to their simultaneous failure. This is known as common cause failure (CCF). In consequence, CCF can be counteracted by implementing redundancy with components technologically diverse; i.e. diversity in redundancy. Notice that whilst identical redundancy aims to address the general problem of random failures, the technique of diversity is used for additionally tackling the particular problem of common cause failures. When implementing diversity, a function is

*Corresponding author. Tel.: +44 7951231263; fax: +44 1142225138.
E-mail address: a.torres@sheffield.ac.uk (A.C. Torres-Echeverría).

0951-8320/$ - see front matter © 2008 Elsevier Ltd. All rights reserved.
doi:10.1016/j.ress.2008.02.010

Nomenclature

RAMS + C	reliability, availability, maintainability and safety plus cost
MOGA	multi-objective Genetic Algorithm
IEC	International Electrotechnical Commission
SIS	safety-instrumented system
LCC	lifecycle cost
N_{ij}	number of ij components
i	subsystem subscript
j	technology kind subscript
PT	pressure measurement/transmitter
TT	temperature measurement/transmitter
FC	final control element (valve and actuator)
PLC	programmable logic controller
CCF	common cause failure
PFD_{avg}	probability of failure on demand (average)
STR	spurious trip rate
SIL	safety integrity level
SFF	safe failure fraction
FT	fault tolerance
TI	test interval
RT	repair time
ε	diagnostic coverage
ε^{S}	diagnostic coverage for safe failures
ε^{D}	diagnostic coverage for dangerous failures
β	beta factor
β^{DD}	beta factor for dangerous detected failures
β^{DU}	beta factor for dangerous undetected failures
β^{SD}	beta factor for safe detected failures
β^{SU}	beta factor for safe undetected failures
C_{MooN}	beta factor modifier for MooN architecture
DI	diversity index
N_s	number of subsystems
C_{ij}	cost item for ij component
C_{PROC}	procurement cost
C_{OP}	operation cost
C_{RISK}	risk cost
C_D	design cost (PLC + instruments design + start upcost)
$C^{purchase}$	purchase cost
$C^{inst/comm}$	installation and commissioning cost
C_{PM}	preventive maintenance cost
C_T	functional test cost
C_{CM}	corrective maintenance (repair) cost
M_{ij}	maintenance frequency of ij component
f_{ij}^{CM}	repair frequency of ij components
C_{STR}	cost of spurious trip rate
C_{SD}	cost of shutdown event
C_{HAZARD}	cost per accident event
SD_{Time}	restart time after shutdown
SD_{LOSS}	cost of loss production per hour
C_{ACC}	cost of accident (assets loss + liability cost)
$F(ACC\|PFD_{avg})$	accident frequency without the SIS per year
R	discount rate
PVF	factor by present value
YL	years life for the system
Sol #	solution number
%Safe	percentage of safe mode failures over total
σ_{share}	fitness sharing parameter (niche size)
σ_{mate}	mating restriction parameter

Failure rates

λ^{T}	total failure rate
λ^{D}	dangerous failure mode
λ^{S}	safe failure mode
λ^{C}	common cause failure mode
λ^{N}	normal (random) failure mode
λ^{DU}	dangerous undetected
λ^{DD}	dangerous detected
λ^{DUN}	dangerous undetected normal (random)
λ^{DUC}	dangerous undetected common cause
λ^{DDN}	dangerous detected normal (random)
λ^{DDC}	dangerous detected common cause
λ^{SU}	safe undetected
λ^{SD}	safe detected
λ^{SUN}	safe undetected normal (random)
λ^{SUC}	safe undetected common
λ^{SDN}	safe detected normal (random)
λ^{SUN}	safe detected common

implemented in different ways in the hope that the same fault is not present in two diverse technologies or techniques.

Allocation of equipment reliability and redundancy has been demonstrated to be a good approach for compliance with IEC 61508 requirements [3]. The present paper is aimed at demonstrating how the consideration of diverse redundancy can provide even more flexibility for RAMS + C-based decision-making as a consequence of its positive impact not only on system safety integrity but also on lifecycle cost.

Reliability optimization by genetic algorithms has been a field of research for about a decade [4–6]. The three main problems addressed by these studies are redundancy allocation, component-reliability allocation and multi-objective reliability optimization [7]. Reliability, availability and cost are the more usual objectives, with constraints of cost and weight [8–10]. The most studied structure is the series–parallel case [5,11]. Additionally, redundant subsystems of n components that require a minimum specified number of k components to work (to be good) for the system to be able to perform its intended function (denominated k-out-of-n:G, where G stands for "good"), have been included in optimization studies by some researchers [6,12]. Other researchers have approached the inclusion of uncertainty [13]. Optimization of protective

systems design with a single-objective genetic algorithm has been carried out by Andrews and Barlett [14] and Pattison and Andrews [15].

Multi-objective optimization of system safety by genetic algorithms was addressed by Martorell et al. [16]. Some other studies completed so far have included safety-related costs or risk-based measures for optimization [8,9,12]. Martorell et al. [17] formulated the problem with the broader concept of reliability, availability, maintainability and safety plus cost (RAMS + C) multi-objective optimization. The inclusion of system safety integrity requirements specified by IEC 61508 in multi-objective optimization was first approached by Torres-Echeverria and Thompson [3]. However, none of the previous studies has addressed explicitly the effect of diverse redundancy for multiple objective optimization of safety related systems design for safety integrity improvement.

The objective of this paper is to explore the effects of using diverse redundancy in SIS design optimization, based on RAMS + C measures. Basically, the problem is formulated as one of redundancy and reliability allocation for a series–parallel system. An application example is provided that illustrates how the proposed approach performs for a chemical reactor safety system for protection against high pressure and temperature. The field instruments subsystems are allowed to choose among several technologically diverse devices, which implements the defence against common cause failures. The results given by the optimizer under these variables are analyzed putting special emphasis on the impact that diversity has in the overall design in terms of the RAMS + C optimized objectives.

2. IEC 61508 safety integrity requirements

It is known that the failure of a safety critical control system to perform its intended function could result in loss of the assets of a company, widespread damage to the environment, harm to personnel and people around the facilities and even loss of life. When safety is achieved by means of the correct operation of a system or equipment, e.g. an SIS, it is said that "functional safety" is being used.

The international standard IEC 61508 "Functional safety of electrical/electronic/programmable electronic safety-related systems" [1] aims to provide a means of ensuring that safety is effectively reached based on functionality of electrical, electronic or programmable electronic systems. Since IEC 61508 is a generic document, non-specific to any industry sector, it is relevant to a big range of different sectors: aircraft industry, process industry, nuclear, automotive, marine, etc.

IEC 61508 comprises seven parts, being a fairly complex standard. However, the first three parts are the basic and most important ones. It can be said that Part 1 addresses management, mainly non-technical, requirements, while Part 2 deals with technical requirements for the hardware realization and Part 3 with technical requirements for software.

The basis of IEC 61508 is the establishment of the safety lifecycle and the definition of safety integrity levels (SIL). The standard requires that every safety function must achieve a specific SIL, determined beforehand based on a previous risk assessment. The SIL is a quantitative index that indicates the acceptable probability of dangerous failure that a system can have to consider it appropriate for a given specific safety integrity requirement. Distinction is made between two different kinds of systems: low-demand mode and high-demand/continuous mode of operation. SISs are usually in low-demand mode of operation. For these, the SIL levels are defined in terms of average probability of failure on demand (Table 1).

In addition to the requirement of bounded PFD_{avg}, the highest SIL that can be claimed for a subsystem's combination for hardware is limited by architectural constraints, which are detailed in IEC 61508 Part 2. These constraints are the hardware fault tolerance and the safe failure fraction (SFF). Table 2 shows the multiple combinations that the standard acknowledges. Notice that subsystems are treated as belonging to two different types: A and B. Subsystems Type A must be composed of components that meet the following conditions: "*their failure modes must be well defined; their behaviour under fault conditions must be completely determined; and sufficient dependable failure data from field (must) show that the claimed rates of failure for (all) dangerous failures are met*" (IEC 61508 Part 2). Practically, a subsystem is considered to be Type B if it does not meet any of these conditions. Goble and Cheddie [18] specify that examples of Type A devices are relays, solenoids, valves and simple electronic modules (e.g. conventional transmitters).

Table 1
SIL for low demand mode of operation (IEC 61508 Part 1)

SIL	PFD_{avg}
4	$\geqslant 10^{-5}$ to $<10^{-4}$
3	$\geqslant 10^{-4}$ to $<10^{-3}$
2	$\geqslant 10^{-3}$ to $<10^{-2}$
1	$\geqslant 10^{-2}$ to $<10^{-1}$

Table 2
SIL architectural constraints (IEC 61508 Part 2)

Subsystem type	A			B		
Hardware FT	0	1	2	0	1	2
SFF (%)						
<60	SIL 1	SIL 2	SIL 3	N/A[a]	SIL 1	SIL 2
60 to <90	SIL 2	SIL 3	SIL 4	SIL 1	SIL 2	SIL 3
90 to <99	SIL 3	SIL 4	SIL 4	SIL 2	SIL 3	SIL 4
⩾99	SIL 3	SIL 4	SIL 4	SIL 3	SIL 4	SIL 4

[a]Not allowed.

Any device with embedded microprocessors (such as smart transmitters) or complex application specific integrated circuits is considered Type B.

The fault tolerance (FT) is the number of faults that a subsystem can tolerate before resulting in system failure. This means that with a fault tolerance of N, $N+1$ faults would cause the system to fail. Notice that the maximum fault tolerance level contemplated in Table 2 is two. This may be considered as a restriction to claim higher SILs for higher FT levels. The effects of this restriction will be seen in the discussion of results in this paper. On the other hand, SFF is the percentage of subsystem failures considered safe: i.e. that they are either safe mode or they are revealed by the diagnostic coverage:

$$\mathrm{SFF} = \frac{(\lambda^{\mathrm{T}} - \lambda^{\mathrm{DU}})}{\lambda^{\mathrm{T}}} \tag{1}$$

If a safety function is implemented using only one single channel the highest SIL level to be claimed is that of the subsystem with the lowest SIL. On the other hand, if a safety function were implemented using multiple channels of subsystems, the SIL would be determined making an analysis of the combination. For example, consider a subsystem of two smart transmitters in parallel, each having a SFF of 92%. Since the overall arrangement would tolerate the failure of one transmitter (still being able to perform its safety function), the fault tolerance is 1. Smart transmitters have a microprocessor embedded, thus classified as Type B devices. Therefore, Table 2 indicates that for a subsystem with Type B devices, fault tolerance of 1 and SFF of 92% a maximum SIL of 3 can be claimed.

Goble and Cheddie [18] substituted the different failure modes rates (see Eqs. (9) and (10)) into Eq. (1), and demonstrated that

$$\mathrm{SFF} = \%\mathrm{Safe} + (1 - \%\mathrm{Safe})\,\varepsilon^{\mathrm{D}} \tag{2}$$

where %Safe is the percentage of only safe mode failures:

$$\%\mathrm{Safe} = \frac{\lambda^{\mathrm{S}}}{\lambda^{\mathrm{T}}} \tag{3}$$

This clearly shows that the SFF depends on the ratio of the safe failure rate to the total failure rate. Note that the SFF is basically the percentage of the total failure rate that does not cause a dangerous undetected failure. Therefore, two components with very different total failure rates (one with very high rate and one with very low total failure rate) can have the same SFF.

Considering this, it is possible to say that the architectural constraints imposed by the IEC 61 508 tables aim to ensure that over-optimistic failure rate data is not used. This could prevent the claim of unrealistically high SIL levels for designs without an adequate level of redundancy. This is permitted by the fact the FT is independent of the total failure rate, and the SFF is a metric of the ratio of failure rates rather than the total.

Determination of the final SIL, which implies calculation of $\mathrm{PFD}_{\mathrm{avg}}$, FT and SFF, requires a sufficient level of modelling detail. This means that the following factors must be included:

- specific architecture (redundancy scheme)
- hardware type: Type A or B
- multiple failure modes (safe and dangerous failure rates)
- diagnostic coverage (share of detected and undetected failures)
- common cause failure (β factor)
- proof test intervals
- repair times.

3. RAMS + C quantification addressing common cause failures

The two dependability objectives to quantify are $\mathrm{PFD}_{\mathrm{avg}}$ and spurious trip rate (STR), in addition to the third objective of life cycle cost (LCC). Quantification of the dependability measures has to address explicitly the effect of diverse redundancy through common cause failure quantification.

There are several methods for quantification of dependability measures, some of them are well-established methods and others recently investigated. Among the former are the well-known techniques of fault tree analysis (FTA) [19], reliability block diagrams (RBD) [20], Markov analysis (MA) [21] and simplified equations (SE) [22]. Hybrid methods pertain to the latter category [23,24]. ISA TR84.0.02 [22] specifically treats the application of SE, FTA and MA to SISs and makes comparison of their modelling capability. Several authors have presented significant descriptions and comparative analysis of the different modelling techniques. Goble [25] analyzed FTA and MA techniques including CCF quantification, concluding that they provide similar results. Goble and Cheddie [18] did it in turn for RBD and FTA. Rouvroye and Brombacher [26] gave an account of FTA, RBD, MA and hybrid methods and contrasted them all exposing their advantages and disadvantages. Bukowski [27] recently made a comparative analysis of MA and SE. Andrews and Ericson [28] compared FTA and MA for several design complexities concluding that FTA delivers the same or very similar results to MA for several cases. Fault trees however are easier to construct and more versatile for modelling large complex systems. Hauge et al. [24] in the PDS Method presents a more refined technique based on calculation formulas, which is presented with an example with simple RBD. The method aims to include failure categories and causes excluded by previous techniques. More recently, Guo and Yang [29] proposed an RBD based approach, which has an equivalent mathematical characteristic to FTA, and addresses and improves the approach originally proposed by IEC 61508 Part 6 based on simplified equations derived through RBD for SIL verification.

In this paper fault trees have been adopted since they can be empowered with house events that confer them the flexibility to accommodate the dynamic changes in redundancy and technology during the optimization process. In addition, a method for common cause failure quantification needs to be accommodated into the fault trees.

Methods for quantification of common cause failures are classified as parametric or non-parametric (stochastic). The main parametric methods for quantification in safety and reliability studies were compiled by Mosleh et al. [30]. Parametric methods can be shock or non-shock models. Shock models include the ones most known for safety and reliability studies: the β factor, the multiple Greek letter and the α factor models. On the other hand, the binomial failure rate model is called a shock model. In general, the simplest and most used method for analysis of safety-related systems is the β factor model. This model is also suggested by IEC 61 508, which presents a practical, more sophisticated version. Lately, the PDS method [24] formulated some further refinements to the basic β factor model.

The basic assumption of the β factor model is that the total failure rate of a component is the sum of independent and common cause failures (Eq. (4)), and that the fraction of the common cause failure rate over the total is the β factor (Eq. (5)):

$$\lambda^{T} = \lambda^{N} + \lambda^{C} \tag{4}$$

$$\beta = \frac{\lambda^{C}}{(\lambda^{N} + \lambda^{C})} \tag{5}$$

These assumptions permit determination of the independent failure fraction (Eq. (6)), called also normal failure, and the common cause failure fraction (Eq. (7)) from the total component failure rate if an estimate of the β factor exists:

$$\lambda^{N} = (1-\beta)\,\lambda^{T} \tag{6}$$

$$\lambda^{C} = \beta\,\lambda^{T} \tag{7}$$

On the other hand, remember that not all hardware failures have the same effects. Dangerous failures have potentially hazardous consequences, i.e. failure to perform the protective function when required; while safe failures result in spurious trips. Consequently, the total failure rate must be split into safe and dangerous failure rates (Eq. (8)). Additionally, the system built-in diagnostic coverage divides both dangerous and safe fractions into those revealed by its action and those not (Eq. (9)). This further splits the total failure rate into detected and undetected failure modes (Eq. (10)):

$$\lambda^{T} = \lambda^{D} + \lambda^{S} \tag{8}$$

$$\lambda^{T} = \varepsilon^{D}\,\lambda^{D} + (1-\varepsilon^{D})\lambda^{D} + \varepsilon^{S}\,\lambda^{S} + (1-\varepsilon^{S})\,\lambda^{S} \tag{9}$$

$$\lambda^{T} = \lambda^{DD} + \lambda^{DU} + \lambda^{SD} + \lambda^{SU} \tag{10}$$

Applying the fractions determined by the β factor (Eqs. (6) and (7)) to total failure rate Eq. (10), we finally include the CCF quantification:

$$\lambda^{T} = \lambda^{DDC} + \lambda^{DDN} + \lambda^{DUC} + \lambda^{DUN} + \lambda^{SDC} + \lambda^{SDN} + \lambda^{SUC} + \lambda^{SUN} \tag{11}$$

The methodology suggested by IEC 61508 Part 6 bases the estimation of the β factor on determining which defence measures against CCF have been put in place, which fall into one of these three categories: reduction of overall failure rates, maximization of independence of channels, and revelation of non-simultaneous CCF. This methodology conveniently extends the basic model so that it can be determined on the basis of the specific system design and plant conditions. It considers the technological differences between logic subsystems and field instrumentation (sensors and final elements) and their dissimilar working conditions. It also takes into account the difference between factors that are enhanced by the diagnostic coverage and those that are not.

The basic β factor model broadly assumes that a common cause failure event will affect all the channels in the redundant subsystem, regardless of the level of redundancy or voting logic. The IEC 61508 methodology fails to incorporate effectively a modification to this assumption so that CCF for different redundant schemes could be better quantified. The PDS Method [24] has recently formulated an extension to the β factor model to correct this situation. Assuming that the probability of simultaneous failures due to a common cause (i.e. the intersection area of independent failure events, Fig. 1) is not the same for different numbers of redundant elements, it enables the β factor to be modified according to the voting logic MooN of the system (where $M<N$, and M out of the N system channels need to vote agree so that the protective function is executed) by a configuration-specific modification factor (Eq. (12)). The PDS method provides values for different voting schemes. For instance, the suggested value for a 1oo3 voting is a third of the intersection for only two components ($C_{\mathrm{MooN}} = 0.3$), shown in Fig. 1.

$$\beta_{\mathrm{MooN}} = \beta \cdot C_{\mathrm{MooN}} \tag{12}$$

Notice that the simplicity of the β factor model reduces the problem of getting CCF data for industrial equipment to the availability of suitable historical statistically

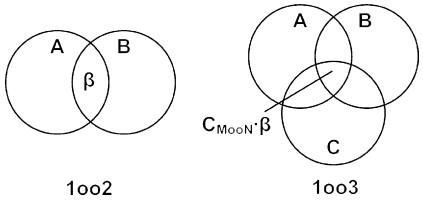


Fig. 1. Modification of the β factor for 1oo3 voting.

significant failure rate data, in addition to the system's design and operation data. Nevertheless, the failure rates for the different failure modes, dangerous and safe, need to be explicitly indicated as well. The same detail is required about the automatic diagnostic coverage per failure mode.

4. Multi-objective optimization by genetic algorithms

4.1. The multi-objective optimization problem

The problem of multi-objective optimization is one of simultaneously minimizing the n different objectives of an objective function vector $\mathbf{f(x)}$:

$$\mathbf{y} = \mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_n(\mathbf{x})) \in \mathbf{Y} \quad (13)$$

where $\mathbf{x}$ is a decision variables vector

$$\mathbf{x} = \{x_1, x_2, \ldots, x_n\} \in \mathbf{X} \quad (14)$$

Usually the objective space $\mathbf{Y}$ is restricted by a vector of constraints

$$\mathbf{g}(\mathbf{x}) = (f_1(\mathbf{x}) \geqslant g_1, f_2(\mathbf{x}) \geqslant g_2, \ldots, f_n(\mathbf{x}) \geqslant g_n) \quad (15)$$

It is generally impossible to find a unique solution that optimizes all the objectives at once. In its place, a set of non-dominated optimal solutions is found, which contain the best solutions in decision space $\mathbf{X}$, called the Pareto optimal set, which constitutes the Pareto front. This implies that a single solution is to be picked up from this set of alternative solutions by the decision maker, which usually is a human.

4.2. Genetic algorithms

Evolutionary computation is a group of techniques that mimic nature's selective process of evolution for solution of optimization problems. Genetic algorithms were initially proposed by Holland [31], and they are one of the most researched of these techniques. They have been widely used for solution of single objective problems. For multi-objective problems, genetic algorithms have been extended to be able to handle two or more objectives, usually in conflict with each other. Concise and comprehensive descriptions of genetic algorithms, especially for RAMS applications, are given in Refs. [32,33].

A genetic algorithm performs a stochastic search guided by the principle of natural selection based on individual genetics. The algorithm simulates the process of evolution of a population of individuals whose genetic characteristics are inherited from those ancestors that were fittest for survival, the same as the natural evolution of species does.

Fig. 2 shows the general scheme of a genetic algorithm. The standard algorithm starts with a random population of several potential solutions in the decision space, denominated "individuals". The individuals are coded representations of the decision variables, usually in binary numbers (although nowadays other codes are used, such as integer and real numbers). These individuals pose a series of attributes that will determine their potential for being an optimal solution of the objectives of the problem at hand. According with its evaluation by an objective function (in the objective space), the individuals are ranked and then assigned a fitness value, which determines their probability of reproduction in the next generation. A fraction of this population is selected for reproduction (recombination), generating a number of "children" called offspring. The offspring are then mutated (commonly flipping one or more bits) and reinserted in the population, usually substituting the same number of individuals. This population represents a new generation. The cycle of evaluation, selection, reproduction and reinsertion is repeated cyclically until a certain condition is achieved. Then the algorithm is stopped. This condition may be the exhaust of a generation count or the compliance with a specific goal.

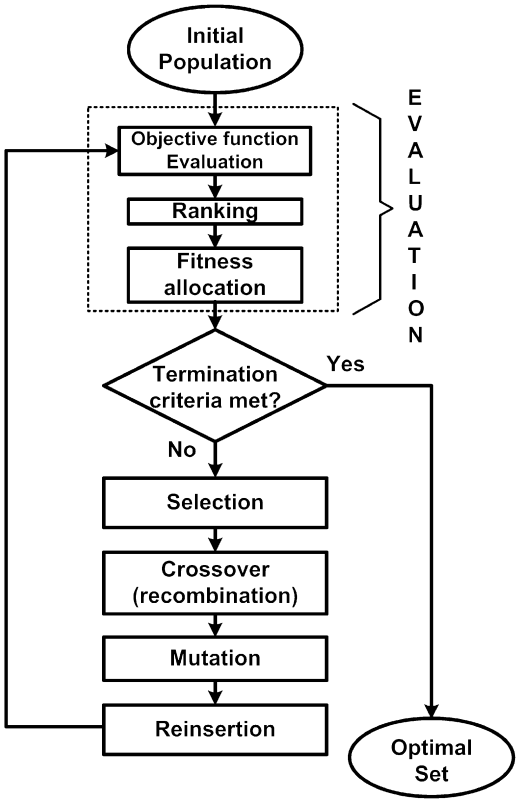


Fig. 2. Standard genetic algorithm flowchart.

4.3. Fonseca and Fleming multi-objective genetic algorithm (MOGA)

The genetic algorithm for solutions of multi-objective problems proposed by Fonseca and Fleming was firstly formulated in [34], and further developed in [35,36]. This algorithm was empowered for multi-objective optimization and named MOGA. The Matlab toolbox used for the present research work is documented in Chipperfield et al. [37]. MOGA was one of the first genetic algorithms to be proposed to approach multi-objective problems based on

the concept of Pareto dominance originally proposed by Goldberg [38].

The Fonseca and Fleming MOGA has the following specific characteristics:

- Ranking based on Pareto dominance.
- Fitness sharing by niching, performed in the objective space.
- Mating restriction.
- Suggested the first truly progressive preference-based method, incorporating the decision maker through goals and priority levels.
- Parallel coordinates for visualization of the trade-off surface in two dimensions.

4.3.1. Evaluation and ranking

Since a cost is calculated for each member of the population, ranking is made after the population is sorted based on the cost (a measure of utility) of each individual.

The Fonseca and Fleming [34] proposition is to assign an individual's rank equal to the number of individuals by which it is dominated. Therefore, the non-dominated solutions have all the same rank. This ranking technique permits to adapt the algorithm for search of the Pareto front in both convex and non-convex trade off surfaces. The original approach ([34]) was to compute the rank:

$$\mathrm{rank}(x_i, t) = p_i^{(t)} + 1 \tag{16}$$

where x_i is an individual at generation t dominated by p_i individuals.

4.3.2. Fitness allocation

Fitness is allocated by previously sorting the population by rank order. Then a mapping with the rank is done (from best to worst ranked individual). This mapping may be linear or exponential. The fitness of individuals with the same rank is then averaged.

The undesired phenomenon of genetic drift is counteracted using fitness sharing. Genetic drift is the effect of the population evolving towards localized regions of the search space, promoting the convergence of the search to a small region of the trade-off surface, which rests efficiency to the GA in the search for the global optima. Although the mutation rate can to some extent counteract this effect when promoting diversity in the population, this must remain between certain low boundaries to avoid converting the evolution process into a simple random search. Through fitness sharing, several individuals close to each other in the objective space have to share resources (i.e. fitness). When sharing the fitness, similar individuals are practically assigned less probability of reproduction, promoting the choice of more diverse characteristics. In other words, similar individuals diminish each other's fitness, and thus their reproducibility. This favours diversification by apportioning the possibility of reproduction to isolated individuals that are equally fit as others. The MOGA has the specific characteristics of implementing the fitness sharing in the objective space. The parameter σ_{share} (niche size) must be calculated for individuals of the same rank, dividing the population up into several niches. The sharing parameter indicates which is the minimum distance between two individuals which does not affect each other's fitness. A niche count is assigned to each individual, set initially to zero and incremented each time another individual meets the requirement for doing so, established by the sharing function ([35]). It is possible to estimate σ_{share} solving

$$N\sigma_{\mathrm{share}}^{n-1} - \frac{\prod_{i=1}^{n}(\Delta_i + \sigma_{\mathrm{share}}) - \prod_{i=1}^{n}\Delta_i}{\sigma_{\mathrm{share}}} = 0 \tag{17}$$

where $\Delta_i = M_i - m_i$, and M and m are the two points that define the parallelogram of the objective space, N is the number of individuals, and n is the number of objectives.

4.3.3. Selection

Stochastic universal sampling (SUS) is used with the aim of achieving the objectives of reducing stochastic selection errors as much as possible, and to reflect through the assigned fitness of individuals their expected eligibility for reproduction. SUS can be seen as a roulette wheel whose slots keep proportion with the fitness of the individuals. Several pointers, which are equally spaced, mark the selection for reproduction once the roulette stopped.

4.3.4. Recombination (crossover)

The mating of individuals from different far apart individuals may create offspring known as lethals: weak individuals with low fitness that diminish the efficiency of the search process. For this reason, to discourage the formation of lethals, mating restriction is introduced based on the distance between selected parents. This simply means that mating restriction promotes the mating of similar individuals (those close to each other in the Pareto set). The main issue introduced here is the selection of the parameter σ_{mate}, which indicates the proximity of individuals to one another be able to mate. The parameter σ_{mate} may be computed the same way as σ_{share}, and in practice is sometimes given the same value.

Mating restriction is implemented as follows: after execution of selection, one individual is chosen, and then another one is picked up whose distance is σ_{mate} and then mated. If such an individual does not exist another one is randomly chosen. The size of the population must be big enough to promote the creation of stable niches, according to the number of decision variables. Additionally, coding must be tailored conveniently to the application. The use of Gray codes may be recommendable for its adjacency property ([34]).

4.3.5. Presentation of results

Fonseca and Fleming propose the use of parallel coordinates as the graphical representation of the trade-

off data, especially for more than two objectives. The method of parallel coordinates is the representation of each non-dominated solution from the Pareto optimal set in two coordinates, being each point in the *X*-axis one of the objectives, plotted against the normalized objectives in the *Y*-axis. Each point is united to one another by a line (each line representing one solution), where lines run non-concurrently they represent non-competing objectives, while crossing lines make evident conflicting objectives.

5. Application problem

The multiple objective optimization by genetic algorithm is applied to the design of a chemical reactor's protection system against high pressure and temperature. This is a series–parallel system. The three objectives to optimize are chosen from RAMS + C measures:

1. Average probability of failure on demand (PFD_{avg}): This measure approaches the system safety integrity, since it represents the probability of loss of safety, and it is the base measure for determining the system's SIL.
2. Spurious trip rate (STR): This is a very relevant measure for industrial systems, since it impacts on losses by deferred production and the trust that the operator places upon the system. If the system has a high nuisance trip frequency, the operator may consider bypassing it, which completely negates the benefit that the system gives for plant safety. Therefore, it is very important to reach a good compromise between STR and PFD_{avg}.
3. Lifecycle cost (LCC): This is an obvious objective itself. The fact that the overall lifecycle cost is encompassed, i.e. procurement, operation and risk costs, contributes to make it congruent with IEC 61508 safety life cycle requirements.

Fig. 3 shows the SIS, which is composed of four subsystems. The field instrumentation comprises both the pressure and temperature sensor/transmitter subsystems, and the final control element subsystem (basically composed of a set of actuator and shutdown valve). The logic solver subsystem is a safety PLC. These subsystems are referred as PT, TT, FC and PLC respectively in the paper. The optimization is focussed on the variability of redundancy and diversity of the field instruments subsystems. It is assumed that the PLC subsystem meets all requirements for at least a SIL 3 application, and its design is not varied during the optimization. The redundancy per subsystem can be varied between one and four channels, and each channel can be chosen from three different technologies in order to implement diversity. These technologies are called A, B and C. They have different failure rates and diagnostic coverage as well as cost specifications. All relevant dependability and lifecycle cost data is presented in Table 3. It is important to bring to the reader's attention that the IEC 61508 device type classification (A or B), and the three different kinds of technologies available per subsystem (named A, B, C), should not be confused. The latter will be referred to as "technology", and the former as "device type".

6. Formulation of quantitative measures

6.1. Quantification of dependability objectives

The two dependability objectives, PFD_{avg} and STR, are quantified based on fault tree analysis. The fault trees are

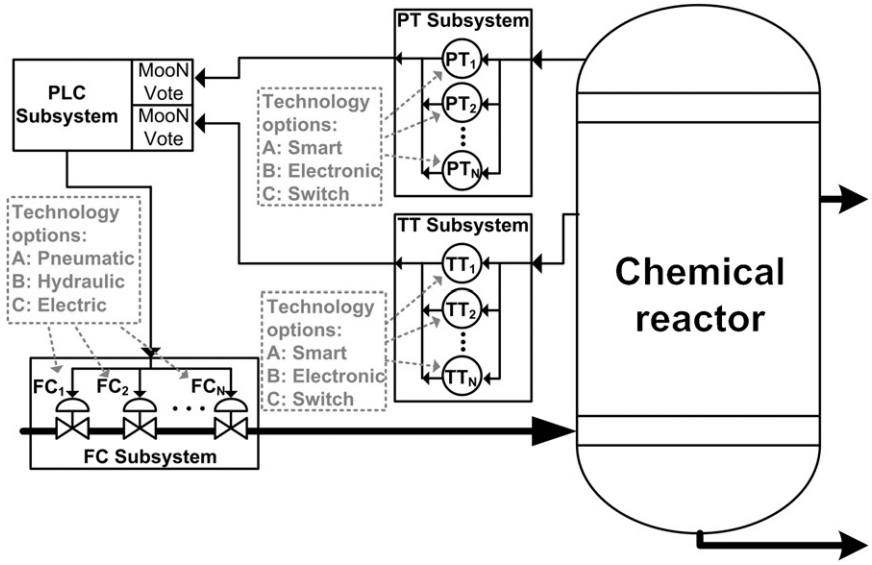


Fig. 3. Chemical reactor protection system.

Table 3
Dependability and lifecycle cost data

Subsystem	Pressure measurement/transmitter			Temperature measurement/transmitter[a]			Final control element (valve & actuator)			Logic solver
Technology	A Smart transmitter	B Conventional electronic transmitter	C Switch	A Smart transmitter	B Conventional electronic transmitter	C Switch	A Air operated[b]	B Hydraulic operated[c]	C Motor operated[c]	Safety PLC
$\varepsilon^S/\varepsilon^D$ (%)	69.2/31.8	56.0/51.1	10/10	97.9/7.5	86.9/45.5	10/10	0/25	0/20	0/10	−/81.25
λ^{SD} ($\times 10^{-6}$/h)	0.265	1.21	0.68	5.05	6.5	0.92	0	0	0	3.46[d]
λ^{SU} ($\times 10^{-6}$/h)	0.118[e]	0.95	6.13	0.11[e]	0.98	8.30	3.94	3.17	9.17	
λ^{DD} ($\times 10^{-6}$/h)	0.048	0.97	0.41	0.026	1.57	0.76	0.84	1.09	0.79	0.026
λ^{DU} ($\times 10^{-6}$/h)	0.103	0.93	3.70	0.322	1.88	6.84	2.51	4.35	7.11	0.006
λ^{T} ($\times 10^{-6}$/h)	0.534	4.06	10.92	5.508	10.93	16.82	7.29	8.61	17.07	3.492
Type	B	A	A	B	A	A	A	A	A	B
SFF (%)	80.7	77.09	66.12	94.15	82.79	59.33	65.57	49.48	58.35	99.83
$C^{purchase}$ ($)	4844	2306	500	2560	1406	500	6940	6400	6200	40000

Lifecycle cost data:
Design/install/commissioning PLC = 10,320 ($)
Repair PLC = 8000 ($/event)
Maintenance PLC = 960 ($/event)
Test PLC = 240 ($/event)
Design overall instrumentation = 3060 ($)
Installation/commissioning per instrument = 600 ($)
Maintenance per instrument = 240 ($/event)
Test per instrument = 60 ($/event)
Repair cost per instrument & PLC = 480 ($/event)
Spares per repair = 7.5% component cost
Loss of production = 2000 ($/h)
Start up cost = 1800 ($)
Catastrophic loss = 150×10^6 ($)
SIS life = 15 (years)
Discount rate = 0.05

Other data:
Repair time = 8 (h)
Shut down time = 24 (h)
Test interval = 1 (year)
Plant risk without SIS = 8.55 ($\times 10^{-3}$/yr)
$C_{1oo2} = 1.0$
$C_{1oo3} = 0.3$
$C_{1oo4} = 0.15$
$\beta = 10\%$[f]

[a]The measurement element is a thermocouple.
[b]Includes the solenoid valve.
[c]Includes the electromechanical relay.
[d]This figure includes $\lambda^{SD} + \lambda^{SU}$.
[e]This failure mode does not cause spurious trip, only quantified for SFF.
[f]Apply for all cases, except for β^{SD} of: PTA = 5%, TTA = 5%, TTB = 2%.

empowered with house events, which confer the fault trees the flexibility to accommodate the dynamic changes in redundancy and technology during the optimization process. Fig. 4 shows the fault tree for quantification of PFD_{avg}, while Fig. 5 corresponds to quantification of STR. The house events permit switching on and off the redundant units per technology per subsystem by changing their logic value to 1 or 0 respectively. Notice that the full fault trees are not displayed. In both fault trees, only the full branch for technology A of the PT subsystem is shown, which is exactly the same for technologies B and C. In the same fashion, the sub-fault trees for the TT and FC subsystems would be identical to the entire PT subsystem sub-tree.

The fault trees are solved by quantification of minimal cut sets. The program developed for quantification of the objective function selects dynamically the cut sets relevant to the candidate solution under evaluation. It then calculates the PFD_{avg} and the STR of this candidate solution. The formulas of the basic events of both fault trees are shown below.

For dangerous independent failure basic events:

$$PFD_{avg} = (1 - C_{MooN}\beta^{DD})\lambda^{DD} RT + (1 - C_{MooN}\beta^{DU})\lambda^{DU}\left(\frac{TI}{2} + RT\right) \quad (18)$$

For dangerous common cause failure:

$$PFD_{avg} = C_{MooN}\beta^{DD}\lambda^{DD} RT + C_{MooN}\beta^{DU}\lambda^{DU}\left(\frac{TI}{2} + RT\right) \quad (19)$$

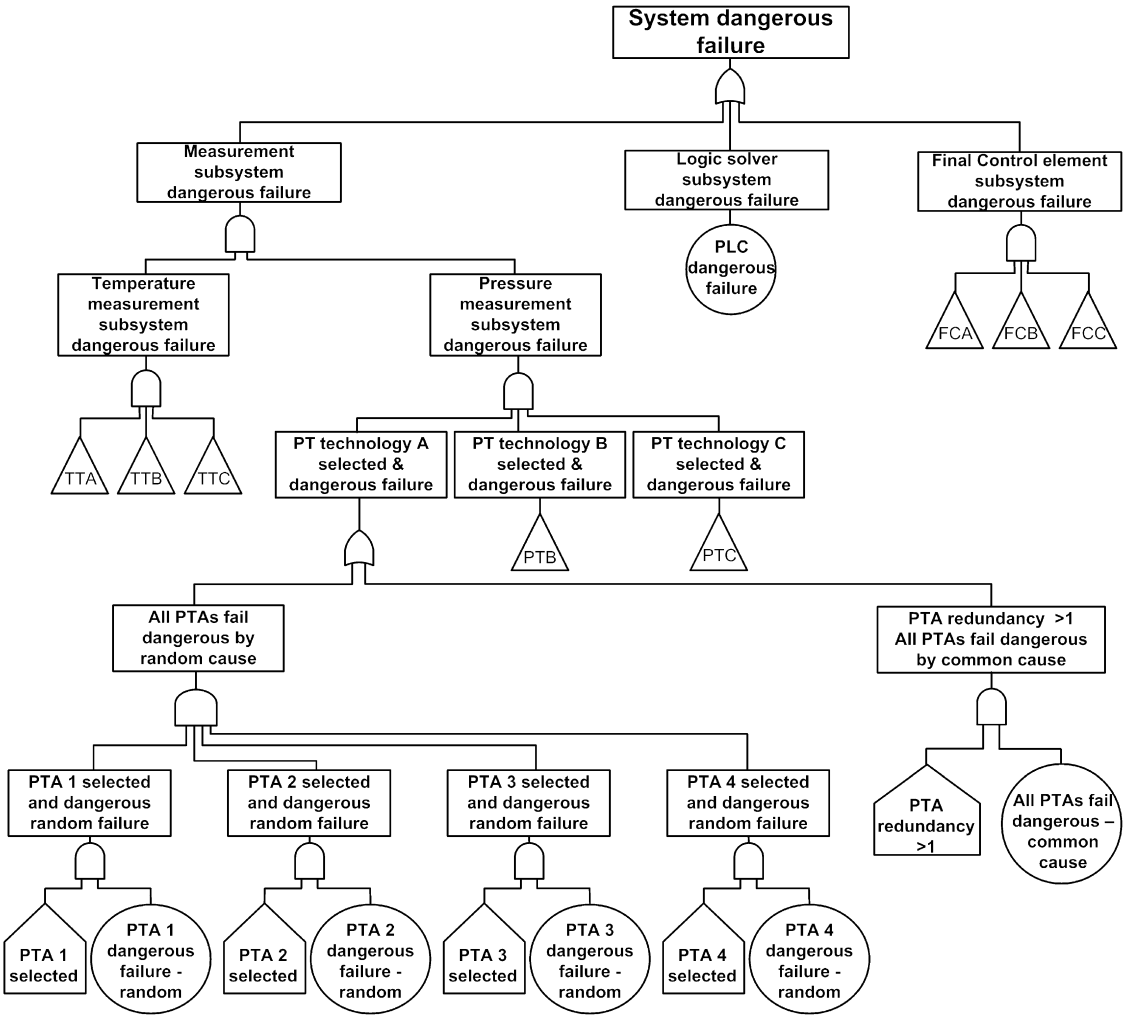


Fig. 4. Fault tree for quantification of probability of failure on demand.

For safe independent failure basic events:

$$\mathrm{STR} = (1 - C_{\mathrm{MooN}}\,\beta^{\mathrm{SD}})\lambda^{\mathrm{SD}} + (1 - C_{\mathrm{MooN}}\,\beta^{\mathrm{SU}})\lambda^{\mathrm{SU}} \quad (20)$$

For safe common cause failure:

$$\mathrm{STR} = C_{\mathrm{MooN}}\,\beta^{\mathrm{SD}}\,\lambda^{\mathrm{SD}} + C_{\mathrm{MooN}}\,\beta^{\mathrm{SU}}\,\lambda^{\mathrm{SU}} \quad (21)$$

The following assumptions apply for the solution of the problem:

- The system operates in low-demand mode.
- Components have constant failure rate.
- Failure of each subsystem is independent from others.
- Once a component has failed it remains in that state until it is repaired.
- Testing and repair are perfect: no faults overlooked, and the item is returned to "as new" condition or normal state after repair.
- The testing strategy is independent test per component.
- For redundancy higher than two only the CCF that affects all redundant components is considered.
- Appropriate measures have been put in place to eliminate coupling among different technologies, so that common cause failure events among them is negligible.

Different β factors for each kind of technology per subsystem have been estimated using the tables presented in IEC 61508 Part 6 Annex D. Specific β factors for each failure mode have been determined, however practically the factor in every case is the same, with some few exceptions for some safe detected failure modes (see the Notes in Table 3).

6.2. Quantification of lifecycle cost

The system LCC is the third objective to optimize, which comprises all procurement, operating and risk costs. The models presented by Goble [25] were revised and tailored to this particular application. The money's time value is considered by calculating the operating and risk costs at present value for the operating life of the system.

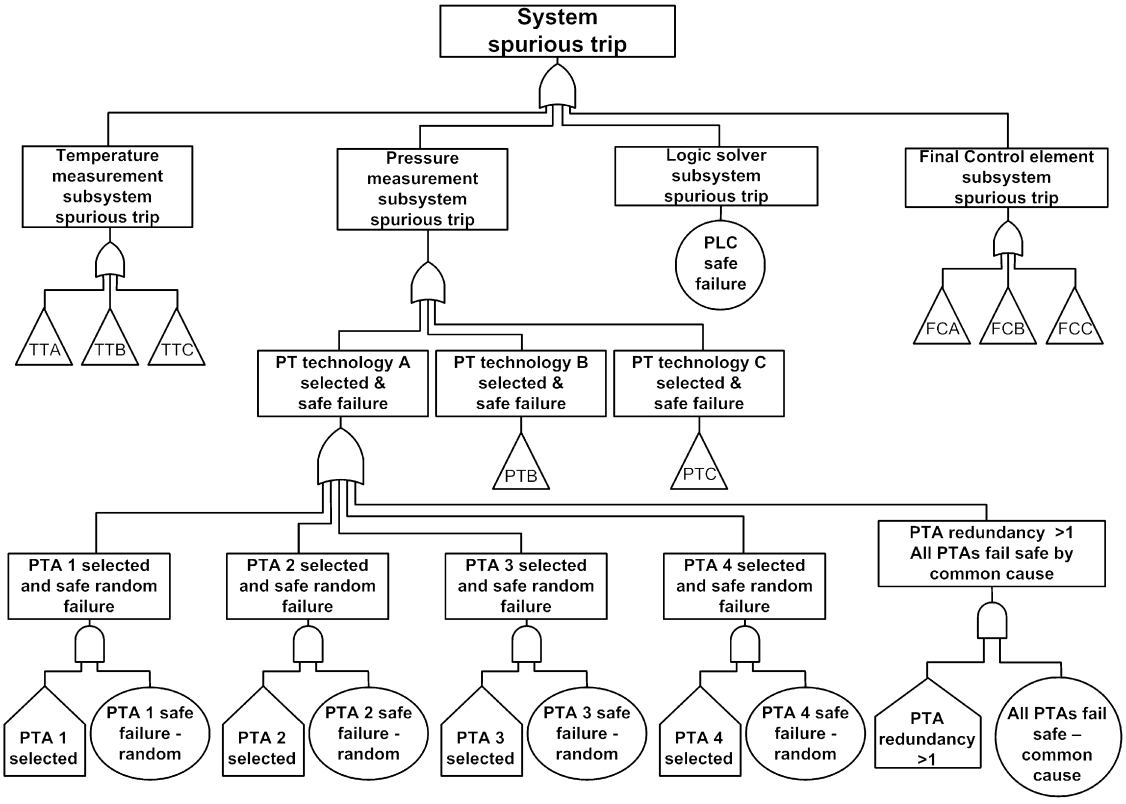


Fig. 5. Fault tree for quantification of spurious trip rate.

The overall lifecycle cost is given by

$$\mathrm{LCC} = C_{\mathrm{PROC}} + (C_{\mathrm{OP}} + C_{\mathrm{RISK}})\,\mathrm{PVF} \tag{22}$$

where the applied factor by present value is

$$\mathrm{PVF} = \frac{1-(1+R)^{-\mathrm{YL}}}{R} \tag{23}$$

Procurement cost per year is quantified as

$$C_{\mathrm{PRO}} = C_{\mathrm{D}} + \sum_{\forall ij}(C_{ij}^{\mathrm{purchase}} + C_{ij}^{\mathrm{inst/comm}})N_{ij} \tag{24}$$

where N_{ij} is the number of components of jth type of technology of the ith subsystem. Basically, these are the decision variables of the problem.

The overall operating cost per year (Eq. (25)) includes preventive maintenance, testing and corrective maintenance. These are respectively calculated in Eqs. (26)–(28):

$$C_{\mathrm{OP}} = C_{\mathrm{PM}} + C_{\mathrm{T}} + C_{\mathrm{CM}} \tag{25}$$

where

$$C_{\mathrm{PM}} = \sum_{\forall ij}\frac{1}{M_{ij}}C_{ij}^{\mathrm{PM}}N_{ij} \tag{26}$$

$$C_{\mathrm{T}} = \sum_{\forall ij}\frac{1}{TI_{ij}}C_{ij}^{\mathrm{T}}N_{ij} \tag{27}$$

$$C_{\mathrm{CM}} = \sum_{\forall ij} f_{ij}^{\mathrm{CM}}\,C_{ij}^{\mathrm{CM}}N_{ij} \tag{28}$$

Being the frequency of repair:

$$f_{ij}^{\mathrm{CM}} = \lambda_{ij}^{\mathrm{Total}} \tag{29}$$

The risk cost per year comprises both the cost per spurious trip rates and the cost of hazard:

$$C_{\mathrm{RISK}} = C_{\mathrm{STR}} + C_{\mathrm{HAZARD}} \tag{30}$$

The cost for spurious trip rate is basically directly proportional to the cost production loss by spurious shutdown:

$$C_{\mathrm{STR}} = \mathrm{STR}\cdot C_{\mathrm{SD}} \tag{31}$$

where

$$C_{\mathrm{SD}} = \mathrm{SD}_{\mathrm{Time}}\cdot \mathrm{SD}_{\mathrm{LOSS}} \tag{32}$$

The cost of hazard includes plant assets loss, deferred production and liability (i.e. fatalities) costs:

$$C_{\mathrm{HAZARD}} = C_{\mathrm{ACC}}\cdot F(\mathrm{ACC}\big|\mathrm{PFD}_{\mathrm{avg}})\cdot \mathrm{PFD}_{\mathrm{avg}} \tag{33}$$

Notice that the $F(\mathrm{ACC}|\mathrm{PFD}_{\mathrm{avg}})$ is the frequency of plant failure (or accident) per year without the SIS, which is actually the frequency of plant failure assuming the SIS is unavailable: i.e. by the Total Probability Theorem, it is the $F(\mathrm{ACC}|\mathrm{PFD}_{\mathrm{avg}})$ term in the expression:

$$F(\mathrm{ACC}) = F(\mathrm{ACC}\big|\mathrm{PFD}_{\mathrm{avg}})\cdot \mathrm{PFD}_{\mathrm{avg}} + F(\mathrm{ACC}\big|1-\mathrm{PFD}_{\mathrm{avg}})\cdot(1-\mathrm{PFD}_{\mathrm{avg}}) \tag{34}$$

6.3. Diversity index

In order to analyze more effectively the effect of diversity on the optimization objective a diversity index was formulated. The objective of this index is to reflect the relative level of both redundancy and diversity to the total number of components. The first formulation made was the following:

$$\mathrm{DI} = \frac{1}{N_s}\sum_{i}^{\mathrm{Ns}} \frac{\text{Technologies}}{\text{Components}} \cdot \frac{\text{Redundancy}}{\text{Components}} \tag{35}$$

where N_s is the number of subsystems. Notice that the term *Redundancy* refers to the number of extra components not necessary for the execution of the safety function but added for the sake of subsystem dependability improvement (i.e. *Components*−1). The term *Technologies* simply refers to the number of different technologies used per subsystem.

This index was somewhat insensitive to the changes in redundancy when this is changed simultaneously with the diversity. For this reason the index was slightly modified resulting in a more sensitive index:

$$\mathrm{DI} = \frac{1}{N_s}\sum_{i}^{N_s} \frac{\text{Technologies} \times \text{Redundancy}}{\text{Components}} \tag{36}$$

7. Implementation of the optimization algorithm

The optimization of the SIS design was based on the Fonseca & Fleming MOGA genetic algorithm [34]. The optimization program was implemented in Matlab, using the MOGA toolbox [37].

The implementation of the optimization algorithm follows the flowchart presented in Fig. 6. For each single subsystem there are 34 unique combinations of redundancy and technologies (making a total of about 40×10^3 possible combinations for the design of the overall system), which were codified and included in a vector. The vector permits access to these combinations by their index. Notice that the decision variables are integer, which favours the use of a binary chromosome. The 21-bits binary chromosome (seven bits per subsystem, as shown in Fig. 7), access the relevant combination in the combinations vector according to the candidate solutions being evaluated. With this information the objective function is called and calculated, which returns a vector with values of the objectives under optimization. These vectors are stored in an Objectives Matrix.

After the genetic operators are applied to the Objectives Matrix, another Best Matrix collects and stores the best-ranked individuals from each generation. Once the program termination criterion is met (i.e. maximum number of generations is exhausted) this matrix is re-ranked, in order to eliminate all the dominated individuals. The resulting pool of solutions constitutes the Pareto Optimal Front.

Tuning was made for the algorithm, and the parameters that provided better results (more diversity and individuals in the Pareto front) were:

- Population: 20 individuals.
- Generations: 300.
- Reproduction and replacement rate: incremental from 10% to 70%.
- Selection algorithm: stochastic universal sampling.
- Crossover mechanism/rate: single point at 70%.
- Mutation mechanism/rate: bit-flipping at 5%.

A second optimization program was created that constrains the design to only variation in redundancy without diversity, with the objective of contrasting the diversity against the homogeneity in the subsystems' design. This means that the level of redundancy can be varied, but all the components of each redundant subsystem must be the same kind of technology.

8. Results

Table 4 presents the Pareto Optimal Front of solutions, which are sorted by decreasing $\mathrm{PFD}_{\mathrm{avg}}$. Each solution is assigned a number, and it shows the number of elements per technology per subsystem (under the headings A, B, C). It presents additionally the optimized objectives per solution and its diversity index (Eq. (36)).

Observe that there is a wealth of solutions in the optimal front that present diversity in redundancy, which are shown by the highlighted sections in the columns PT, TT and FC of the table. Specifically, the majority of solutions present diversity in the FC subsystem (84%). Additionally, 48% and 23% of the PT, TT subsystems respectively are also diverse. With this, it can be said the optimizer favours the implementation of diversity for the optimal solutions. Additionally, it is notable that diverse redundancy is presented more frequently in solutions with higher SIL. Notice how the diversity (highlighted areas) in the subsystems increases along with the SIL figures.

Fig. 8 serves a dual purpose. First of all, it presents the graphs for the obtained optimal solutions of Table 4 (optimization with diversity). Secondly, it shows over the same graphs the optimal solution obtained from the optimization without diversity, enabling the reader to compare both diverse and non-diverse approaches. One can notice that there exist discontinuities in the search space, presumably caused by the discrete nature of the problem. Analyzing Table 4 it is possible to find an explanation. "Jumps" are caused when both redundancy and diversity are changed simultaneously between two consecutive solutions, creating clusters of solutions and empty spaces in the search space and the Pareto optimal front.

The graphs in Fig. 8 illustrate the most relevant demonstration of this study. Notice that, when the non-diverse solutions and the diverse ones do not coincide, the

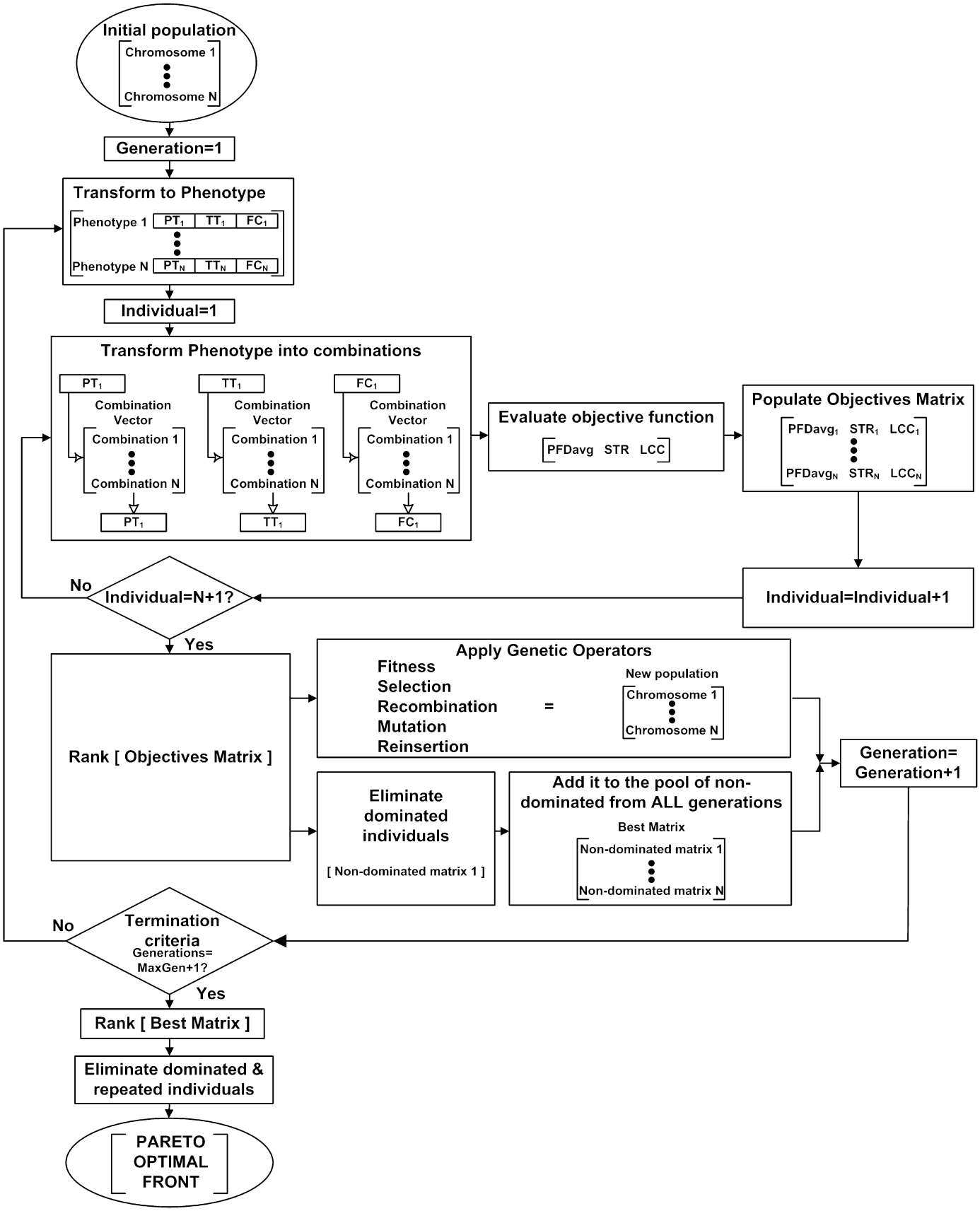


Fig. 6. Flowchart of the implementation of the optimization algorithm.

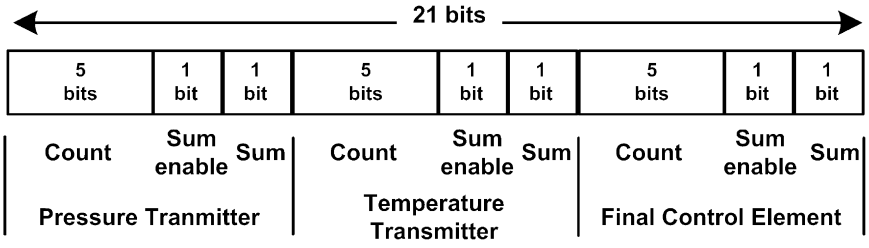


Fig. 7. Chromosome code for the SIS design.

non-diverse are dominated by the diverse solutions. This leads us to conclude that, in general, the optimization permitting diversity gives better solutions than the one without diversity.

The parallel coordinates graph is presented in Fig. 9. This graph shows the optimal solutions with their optimized objectives in normalized parallel coordinates. Observe that the PFD_{avg} is in logarithmic scale, and that the SIL achieved by this figure may be different from the final SIL (further commented below). On the other hand, crossing lines indicate that the PFD_{avg} is generally in conflict with both the STR and the LCC. Nevertheless, there are some exceptions to this observation for SIL levels 1 and 2, where PFD_{avg} and LCC are sometimes harmonious (non-conflicting) objectives, better observed in Fig. 10.

The diversity index plotted against the three optimized objectives per optimal solution is presented in Fig. 10. Recall that the diversity index aims to reflect the relative level of diversity in the overall system. The graph shows a general trend where the larger the diversity index the lower PFD_{avg} and the higher STR and LCC. This trend is, however, somewhat different in solutions with low levels of redundancy (SIL 1 and 2 determined by PFD_{avg}, see the corresponding column in Table 4 as well). In solutions 1–10 some simultaneous changes in redundancy and diversity cause the PFD_{avg} and LCC to become harmonizing objectives, and thus a higher DI represents a lower both PFD_{avg} and LCC. The general trend leads us to think that the higher levels of diversity benefit the system safety integrity, making possible to achieve a design with lower levels of PFD_{avg}.

9. Discussion

Analyzing the data provided in Table 4 it can be seen that some technologies were much less competent for the optimal design. Specifically, observe technologies C for the two transmitter subsystems. These appear only in a very few solutions, and it may be said that only when it is very necessary to improve diversity (solutions 56–64). In contrast, smart transmitters appear in all the optimal solutions, which indicate that they are preferred by the optimizer. Remember that smart transmitters are classified as Type B devices (IEC 61508 classification), and for these devices higher levels of fault tolerance are required to achieve the same SIL levels that Type A devices do (see Table 2). Despite this restriction, smart transmitters are still used in all the 64 solutions given by the optimizer. This suggests that for high dependability requirements quality (low failure rates and high diagnostic coverage) outweighs cost.

The case of temperature transmitters is particularly notable. The temperature and pressure transmitter subsystems are in parallel, eliminating the need for implementation of high levels of redundancy in both of them (see solutions 1–46 in Table 4). Temperature transmitters have redundancy higher than two only in solutions with very low PFD_{avg} values. It is evident that the optimizer gives preference to the PTs over the TTs, even thought TTs have lower purchase costs. Notice that TTs have higher total failure rates than PTs with the same technology, particularly higher safe failure rates. This impacts negatively on the cost of losses by higher spurious trip rates. It can be said that the dominance of these costs exerts considerable influence on the optimizer so that the optimizer discards the TTs as a better choice over the PTs.

Table 4 presents two columns with SIL values. The first one is the SIL that the systems would achieve according to its PFD_{avg} (according with Table 1). The second value is the final SIL, which takes into account the architectural constraints. Observe that in most of the solutions the final SIL is limited by the architectural constraints to a lower level than the one corresponding to their PFD_{avg}. This suggests that for these solutions better levels of SFF and FT are required to make them congruent with the PFD_{avg}. This may be interpreted as the failure rates presented by the equipment may be overoptimistic. Nevertheless, it is important to highlight that IEC 61508 only indicates the SIL for up to a maximum fault tolerance of two (Table 2). This has been interpreted as a restriction on taking credit for a larger SIL with higher FT. It is worth mentioning that if this were permitted, more solutions in Table 4 would have a higher final SIL. Specifically, solutions 59 and from 61 to 64 would achieve SIL 4.

Analyzing the data presented in Table 4 it is possible to find many examples that reinforce the hypothesis previously made from Fig. 10 that increasing diversity reduces the PFD_{avg}. The diversity index of several solutions is highlighted in the last column of the table. As explained next, among these solutions several pairs can be made for which the diversity index is consistently larger for more

Table 4
Optimization results with diversity index

Sol #	Subsystem									Results					
	PT			TT			FC			PFD_{avg}	STR (/year)	SIL PFD_{avg}	SIL final	LCC ($\times 10^5$ $)	DI
	A	B	C	A	B	C	A	B	C						
1	1	0	0	1	0	0	0	1	0	$1.91236952329 \times 10^{-2}$	0.105	1	1	3.983	0.00
2	2	0	0	1	0	0	0	1	0	$1.91231201575 \times 10^{-2}$	0.107	1	1	4.080	0.17
3	3	0	0	1	0	0	0	1	0	$1.91230751771 \times 10^{-2}$	0.109	1	1	4.178	0.22
4	1	0	0	1	0	0	1	0	0	$1.10477752329 \times 10^{-2}$	0.111	1	1	2.978	0.00
5	2	0	0	1	0	0	1	0	0	$1.10472001575 \times 10^{-2}$	0.114	1	1	3.075	0.17
6	3	0	0	1	0	0	1	0	0	$1.10471551771 \times 10^{-2}$	0.116	1	1	3.173	0.22
7	1	0	0	1	0	0	0	2	0	$2.23221566451 \times 10^{-3}$	0.130	2	1	1.936	0.17
8	2	0	0	1	0	0	0	2	0	$2.23164058915 \times 10^{-3}$	0.132	2	2	2.033	0.33
9	3	0	0	1	0	0	0	2	0	$2.23159560875 \times 10^{-3}$	0.134	2	2	2.131	0.39
10	4	0	0	1	0	0	0	2	0	$2.23158602014 \times 10^{-3}$	0.137	2	2	2.228	0.42
11	1	0	0	1	0	0	1	1	0	$2.37630341174 \times 10^{-4}$	0.139	3	1	1.754	0.33
12	2	0	0	1	0	0	1	1	0	$2.37055265814 \times 10^{-4}$	0.141	3	2	1.851	0.50
13	3	0	0	1	0	0	1	1	0	$2.37010285417 \times 10^{-4}$	0.144	3	2	1.949	0.56
14	4	0	0	1	0	0	1	1	0	$2.37000696805 \times 10^{-4}$	0.146	3	2	2.047	0.58
15	1	1	0	1	0	0	1	1	0	$2.36993721879 \times 10^{-4}$	0.158	3	2	1.920	0.67
16	3	1	0	1	0	0	1	1	0	$2.36991186720 \times 10^{-4}$	0.163	3	2	2.115	0.83
17	1	0	0	1	0	0	1	2	0	$5.14761014431 \times 10^{-5}$	0.164	4	1	1.931	0.44
18	2	0	0	1	0	0	1	2	0	$5.09010260834 \times 10^{-5}$	0.166	4	2	2.028	0.61
19	3	0	0	1	0	0	1	2	0	$5.08560456863 \times 10^{-5}$	0.169	4	2	2.125	0.67
20	1	0	0	1	0	0	2	1	0	$5.00994103623 \times 10^{-5}$	0.170	4	1	1.997	0.44
21	2	0	0	1	0	0	2	1	0	$4.95243350025 \times 10^{-5}$	0.172	4	2	2.094	0.61
22	3	0	0	1	0	0	2	1	0	$4.94793546054 \times 10^{-5}$	0.175	4	2	2.192	0.67
23	4	0	0	1	0	0	2	1	0	$4.94697659932 \times 10^{-5}$	0.177	4	2	2.290	0.69
24	1	1	0	1	0	0	2	1	0	$4.94627910677 \times 10^{-5}$	0.189	4	2	2.163	0.78
25	2	1	0	1	0	0	2	1	0	$4.94604398146 \times 10^{-5}$	0.191	4	2	2.260	0.89
26	1	0	0	1	0	0	1	3	0	$3.35589320790 \times 10^{-5}$	0.193	4	1	2.149	0.50
27	1	0	0	1	0	0	2	2	0	$2.98222459436 \times 10^{-5}$	0.195	4	1	2.196	0.50
28	2	0	0	1	0	0	2	2	0	$2.92471705838 \times 10^{-5}$	0.197	4	2	2.293	0.67
29	3	0	0	1	0	0	2	2	0	$2.92021901867 \times 10^{-5}$	0.200	4	2	2.390	0.72
30	4	0	0	1	0	0	2	2	0	$2.91926015744 \times 10^{-5}$	0.202	4	2	2.488	0.75
31	1	1	0	1	0	0	2	2	0	$2.91856266489 \times 10^{-5}$	0.214	4	2	2.362	0.83
32	2	1	0	1	0	0	2	2	0	$2.91832753958 \times 10^{-5}$	0.216	4	2	2.459	0.94
33	3	1	0	1	0	0	2	2	0	$2.91830914890 \times 10^{-5}$	0.219	4	2	2.557	1.00
34	2	2	0	1	0	0	2	2	0	$2.91830401816 \times 10^{-5}$	0.233	4	2	2.616	1.00
35	1	0	0	1	0	0	2	1	1	$2.78905818210 \times 10^{-5}$	0.251	4	1	2.451	0.75
36	2	0	0	1	0	0	1	2	1	$2.73584661064 \times 10^{-5}$	0.247	4	2	2.482	0.92
37	2	0	0	1	0	0	2	1	1	$2.73155064612 \times 10^{-5}$	0.253	4	2	2.548	0.92
38	3	0	0	1	0	0	1	2	1	$2.73134857093 \times 10^{-5}$	0.249	4	2	2.579	0.97
39	4	0	0	1	0	0	1	2	1	$2.73038970971 \times 10^{-5}$	0.251	4	2	2.677	1.00
40	1	1	0	1	0	0	1	2	1	$2.72969221716 \times 10^{-5}$	0.263	4	2	2.551	1.08
41	3	0	0	1	0	0	2	1	1	$2.72705260641 \times 10^{-5}$	0.255	4	2	2.646	0.97
42	4	0	0	1	0	0	2	1	1	$2.72609374519 \times 10^{-5}$	0.257	4	2	2.744	1.00
43	1	1	0	1	0	0	2	1	1	$2.72539625264 \times 10^{-5}$	0.269	4	2	2.617	1.08
44	2	1	0	1	0	0	2	1	1	$2.72516112733 \times 10^{-5}$	0.272	4	2	2.714	1.19
45	3	1	0	1	0	0	2	1	1	$2.72514273664 \times 10^{-5}$	0.274	4	2	2.812	1.25
46	2	2	0	1	0	0	2	1	1	$2.72513760591 \times 10^{-5}$	0.289	4	2	2.871	1.25
47	2	1	0	2	0	0	2	1	1	$2.72513754906 \times 10^{-5}$	0.315	4	3	2.996	1.36
48	1	1	0	1	1	0	2	1	1	$2.72513705522 \times 10^{-5}$	0.335	4	2	3.001	1.42
49	3	1	0	2	0	0	2	1	1	$2.72513568894 \times 10^{-5}$	0.317	4	3	3.094	1.42
50	2	2	0	2	0	0	2	1	1	$2.72513516999 \times 10^{-5}$	0.332	4	3	3.153	1.42
51	2	1	0	1	1	0	2	1	1	$2.72513511261 \times 10^{-5}$	0.337	4	2	3.098	1.53
52	3	1	0	1	1	0	2	1	1	$2.72513496067 \times 10^{-5}$	0.340	4	2	3.196	1.58
53	2	2	0	1	1	0	2	1	1	$2.72513491828 \times 10^{-5}$	0.354	4	2	3.255	1.58
54	2	1	0	2	1	0	2	1	1	$2.72513491781 \times 10^{-5}$	0.381	4	3	3.380	1.64
55	3	1	0	2	1	0	2	1	1	$2.72513490244 \times 10^{-5}$	0.383	4	3	3.478	1.69
56	2	1	1	1	1	0	2	1	1	$2.72513489941 \times 10^{-5}$	0.397	4	2	3.442	1.83
57	2	2	0	1	2	0	2	1	1	$2.72513489828 \times 10^{-5}$	0.416	4	3	3.620	1.69
58	3	1	0	1	1	1	2	1	1	$2.72513489783 \times 10^{-5}$	0.420	4	3	3.648	1.92
59	3	1	0	3	1	0	2	1	1	$2.72513489783 \times 10^{-5}$	0.428	4	3	3.766	1.75
60	2	1	1	2	1	0	2	1	1	$2.72513489625 \times 10^{-5}$	0.440	4	3	3.724	1.94
61	2	1	1	3	1	0	2	1	1	$2.72513489599 \times 10^{-5}$	0.485	4	3	4.013	2.00
62	2	2	0	1	2	1	2	1	1	$2.72513489596 \times 10^{-5}$	0.497	4	3	4.072	2.00
63	1	2	1	2	2	0	2	1	1	$2.72513489593 \times 10^{-5}$	0.517	4	3	4.149	2.00
64	2	1	1	2	1	1	2	1	1	$2.72513489590 \times 10^{-5}$	0.521	4	3	4.177	2.25

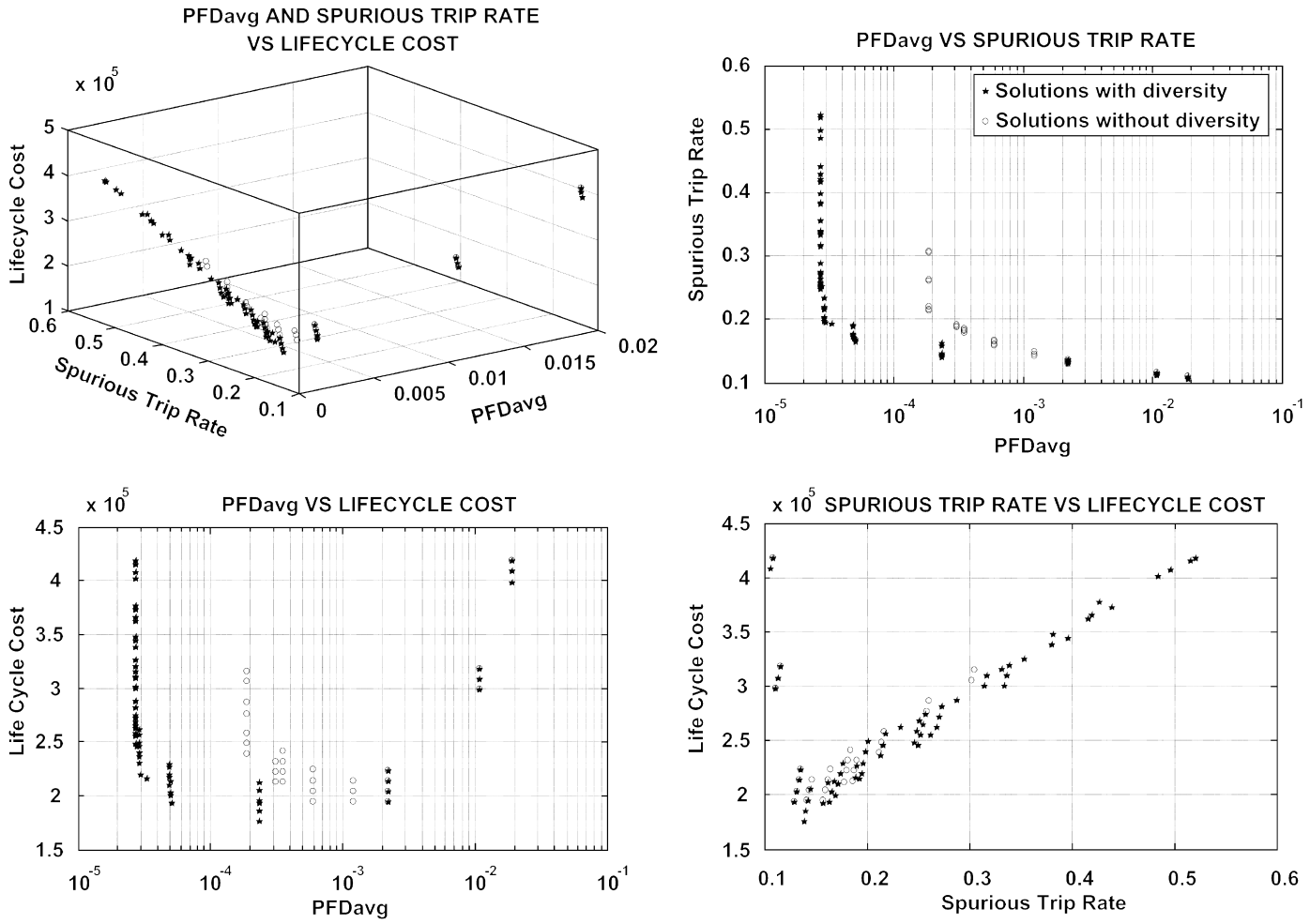


Fig. 8. Comparison of diverse vs non-diverse optimal solutions.

diverse solutions, and which shows improvements in the PFD_{avg}. Three different cases have been extracted:

(a) The following pairs have exactly the same level of redundancy with different level of diversity: 27–35, 28–36, 29–38, 29–41, 30–39, 30–42, 31–40, 31–43, 32–44, 33–45, 34–46, 47–51, 50–53, 55–58, 57–62, 61–64, 53–56 and 55–60. Observe that the solutions with higher diversity have lower PFD_{avg}.
(b) For the pairs 7–11, 8–12, 9–13 and 10–14, with same redundancy and different diversity as well, the sole increment of diversity produces the simultaneous reduction of not only PFD_{avg} but LCC as well.
(c) Finally, for the adjoining pairs 14–15, 23–24, 30–31, 39–40 and 42–43 the PFD_{avg} is reduced by reducing redundancy and increasing diversity simultaneously, which also results in lower LCC.

In conclusion, increasing the diversity enhances the system's PFD_{avg}, and in some cases it may even improve the LCC.

A word of caution is necessary about the search of lower levels of PFD_{avg}. Albeit the general increment of the diversity index generally produces decreasing PFD_{avg}, the overall benefit obtained must be carefully considered. Fig. 8 shows that some marginal improvements in PFD_{avg} can incur in very large increments in LCC and STR. Due to this situation, solutions 40–64 may not be useful, unless very specific requirements were being sought (e.g. either SIL 3 by architectural constraints or to achieve a specific PFD_{avg} value), such as for example the ones adopted in the nuclear energy field.

10. Concluding remarks

This paper integrates the multi-objective optimization of SISs based on RAMS + C measures in compliance with the requirements imposed by the international standard IEC 61508. Particularly, it puts emphasis on the study of the impact of introducing diversity into the redundant subsystems upon the SIS design optimization. Looking for a practical approach, the optimization methodology has been applied to a safety function for a chemical reactor. The level of modelling detail and the assessment of functional system safety necessary to meet IEC 61508 provisions are analyzed, and further applied to

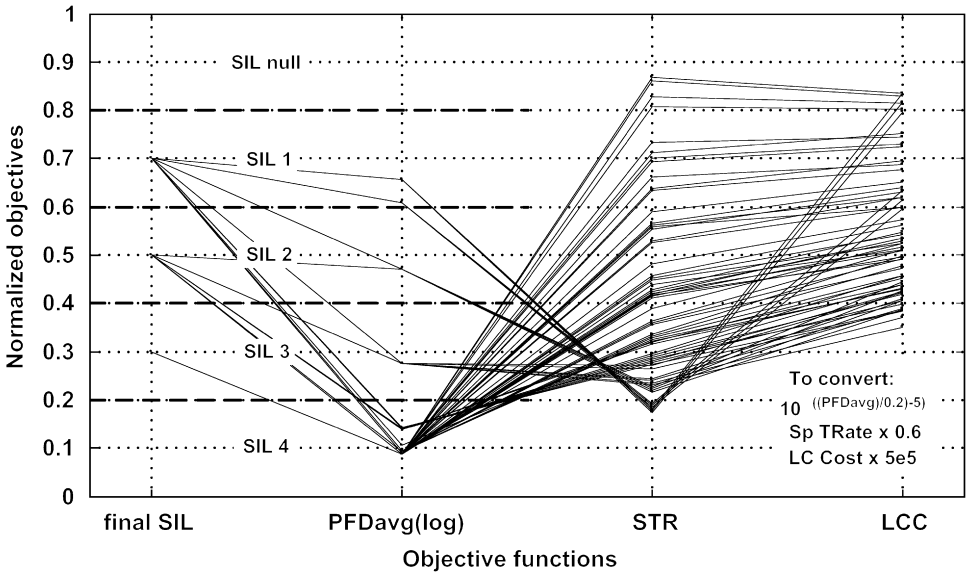


Fig. 9. Parallel coordinates graph of the optimal solution.

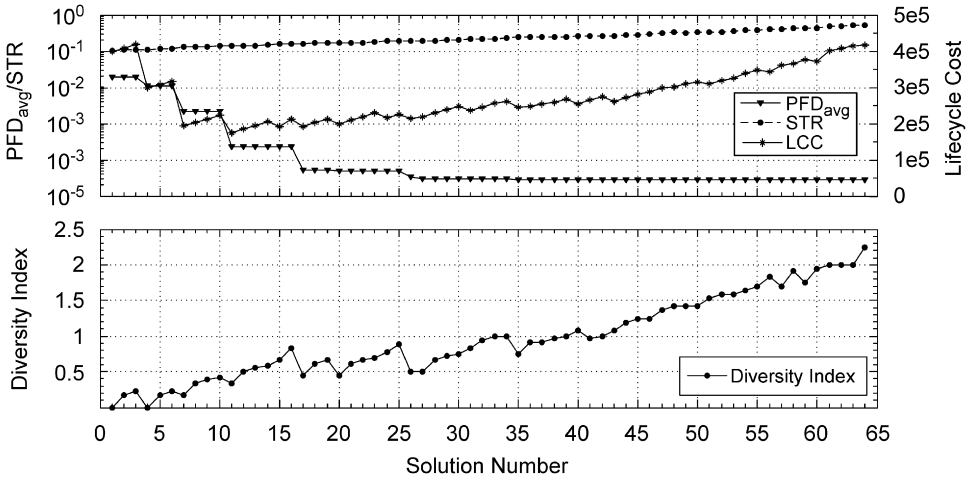


Fig. 10. Comparison of diversity index vs optimized objectives.

the optimization process. The overall methodology presented makes the contribution of this article highly relevant for practical industrial applications.

The use of redundancy to enhance the design of SISs introduces the issue of common cause failure. The effects of CCF can be so substantial that it may even dominate dependability calculations, such as probability of failure on demand. This has a direct negative impact on the safety integrity level achieved by the system. The results presented here show that, in general, the design optimization with diversity gives better results than the one without diversity. The results with diversity generally dominates the ones without diversity, which means that the former achieve better trade-off between the optimized objectives.

On the other hand, the set of optimal solutions obtained showed that introducing diversity as a defence measure against CCF has a positive impact on the functional safety of the safety system, which in turn improves the overall plant safety. Specifically, it has been seen that sometimes the sole increment of diversity, even without altering system redundancy, reduces the overall PFD_{avg}. However, in contrast, this gain in safety integrity usually has an increment in some costs and the spurious trip rate, which may have a negative impact in the overall system's lifecycle cost and in the trust given to the SIS performance, justified only by the specific safety requirements of the plant.

Generally speaking, this study is a further development of the previous work done by Torres-Echeverria and Thompson [3], with the aim of developing a complete methodology optimization of SIS design in harmony with IEC 61508 requirements. Further future research is planned to investigate more complex architectures and to accommodate maintainability optimization options, so as to make it extensive to the most relevant issues faced for design and operation of industrial SIS.

References

[1] International Electrotechnical Commission. IEC 61508 Functional safety of electrical/electronic/programmable electronic safety-related systems. Parts 1–7, Switzerland, 1998–2005.

[2] International Electrotechnical Commission. IEC 61511 Functional safety—safety instrumented systems for the process industry sector. Parts 1–3, Switzerland, 2003.

[3] Torres-Echeverria AC, Thompson HA. Multi-objective genetic algorithm for optimization of system safety and reliability based on IEC 61508 requirements: a practical approach. Proc IMechE PtO: J Risk Reliab 2007;221(O3):193–205.

[4] Ida K Gen M, Yokota T. System reliability optimization problems with several failure modes by genetic algorithm. In: Proceedings of the 16th international conference on computers and industrial engineering. Ashikaga, Japan, 1994. p. 349–52.

[5] Painton L, Campbell J. Genetic algorithms in optimization of system reliability. IEEE Trans Reliab 1995;44(2):172–8.

[6] Coit DW, Smith A. Reliability optimization of series–parallel systems using a genetic algorithm. IEEE Trans Reliab 1996;45(2): 254–60.

[7] Kuo W, Prasad VR. An annotated overview of system-reliability optimization. IEEE Trans Reliab 2000;49(2):176–87.

[8] Yang J, Hwang M, Sung T, Jin Y. Application of genetic algorithm for reliability allocation in nuclear power plants. Rel Eng Syst Saf 1999;65(3):229–38.

[9] Busacca PG, Marseguerra M, Zio E. Multiobjective optimization by genetic algorithms: application to safety systems. Rel Eng Syst Saf 2001;72(1):59–74.

[10] Salazar D, Rocco CM, Galván BJ. Optimization of constrained multiple-objective reliability problems using evolutionary algorithms. Rel Eng Syst Saf 2006;91(9):1057–70.

[11] Elegbede C, Adjallah K. Availability allocation to repairable systems with genetic algorithms: a multi-objective formulation. Rel Eng Syst Saf 2003;82(3):319–30.

[12] Cantoni M, Marseguerra M, Zio E. Genetic algorithms and Monte Carlo simulation for optimal plant design. Rel Eng Syst Saf 2000; 68(1):29–38.

[13] Marseguerra M, Zio E, Podofillini L. Optimal reliability/availability of uncertain systems via multi-objective genetic algorithms. IEEE Trans Reliab 2004;53(3):434–524.

[14] Andrews JD, Bartlett LM. Genetic algorithm optimisation of a firewater deluge system. Qual Rel Eng Int 2003;19(1):39–52.

[15] Pattison RL, Andrews JD. Genetic algorithms in optimal safety system design. Proc IMechE E: J Proc Mech Eng 1999;213:187–97.

[16] Martorell S, Sánchez A, Carlos S, Serradel V. Alternatives and challenges in optimizing industrial safety using genetic algorithms. Rel Eng Syst Saf 2004;86(1):25–34.

[17] Martorell S, Villanueva JF, Carlos S, Nebot Y, Sánchez A, Pitchard JL, et al. RAMS+C informed decision-making with application to multi-objective optimization of technical specifications and maintenance using genetic algorithms. Rel Eng Syst Saf 2005;87(1): 65–75.

[18] Goble WM, Cheddie H. Safety instrumented systems verification. USA: The Instrumentation, Systems and Automation Society; 2005.

[19] International Electrotechnical Commission. IEC 1025 Fault Tree Analysis. Switzerland, 1990.

[20] International Electrotechnical Commission. IEC 1078 Analysis techniques for dependability—reliability block diagrams. Switzerland, 1991.

[21] International Electrotechnical Commission. IEC 1165 Application of Markov techniques. Switzerland, 1995.

[22] The Instrumentation, Systems and Automation Society. ISA TR84.0.02 Safety Instrumented Systems. Safety integrity level evaluation techniques. Draft Version 5. Parts 1–5, USA, 1999.

[23] Knegtering B, Brombacher A. Application of micro Markov models for quantitative safety assessment to determine safety integrity levels as defined by IEC 61508 standard for functional safety. Rel Eng Syst Saf 1999;66(2):171–5.

[24] Hauge S, Hokstad P, Langseth H, Oien K. Reliability prediction method for safety instrumented systems. PDS method handbook 2006 edition. Norway: SINTEF; 2006.

[25] Goble WM. Control systems safety evaluation & reliability. USA: The Instrumentation, Systems and Automation Society; 1998.

[26] Rouvvroye J, Brombacher A. New quantitative safety standards; different techniques, different results? Rel Eng Syst Saf 1999;66(2): 121–5.

[27] Bukowski J. A comparison of techniques for computing PFD average. In: Proceedings of the 51st Annual reliability and maintainability Symposium. Alexandria, VA, USA, 2005. p. 590–595.

[28] Andrews JD, Ericson II CA. Fault tree and Markov analysis applied to various design complexities. In: Proceedings of the 18th international system safety conference. Texas, USA, 2000.

[29] Guo H, Yang X. A simple reliability block diagram method for safety integrity verification. Rel Eng Syst Saf 2007;92(9):1267–73.

[30] Mosleh A, Fleming KN, Parry GW, Paula HM, Worledge DH, Rasmuson DM. Procedures for treating common cause failures in safety and reliability studies. NUREG/CR-4780 (EPRI NP-5613). USA: Nuclear Regulatory Commission; 1988.

[31] Holland JH. Adaptation in natural and artificial systems. USA: University of Michigan Press; 1975.

[32] Marseguerra M, Zio E, Martorell S. Basics of genetic algorithms optimization for RAMS applications. Rel Eng Syst Saf 2006;91(9): 977–91.

[33] Konak A, Coit DW, Smith AE. Multi-objective optimization using genetic algorithms: a tutorial. Rel Eng Syst Saf 2006;91(9):992–1007.

[34] Fonseca CM, Fleming PJ. Genetic algorithms for multi-objective optimization: formulation, discussion and generalisation. In: Proceedings of the fifth international conference on genetic algorithms. San Mateo, CA, USA, 1993. p. 416–23.

[35] Fonseca CM, Fleming PJ. Multiobjective genetic algorithms made easy: selection, sharing and mating restriction. In: Proceedings of the first international conference on genetic algorithms in engineering systems: innovations and application (GALESIA '95). Stevenage, UK, 1995. p. 45–52.

[36] Fonseca CM, Fleming PJ. Multiobjective optimization and multiple constraint handling with evolutionary algorithms I: a unified formulation. IEEE Trans Syst Man Cybernet A: Syst Hum 1998;28(1): 38–47.

[37] Chipperfield A, Fleming P, Polheim H, Fonseca C. Genetic algorithm toolbox user's guide. Research report 512. UK: University of Sheffield; 1994.

[38] Goldberg DE. Genetic algorithms in search, optimization and machine learning. Reading, MA: Addison-Wesley; 1989.